

## SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA07

### COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.  
(BL3)

Tool Weightage: 30%

duct:

19256504 (Name / Reg No) certify that this submission is my original  
that I have adhered to the guidelines specified for this assessment.  
that any violation of academic integrity rules will result in disciplinary

the student: Tharigai S.

| Criteria                     | Weightage (%) | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks | Q5. |
|------------------------------|---------------|------------|------------|------------|------------|-----|
| Understanding of Scenario    | 20%           | 1          | 1          | 1          | 1          | 1   |
| Identification of Key Issues | 20%           | 1          | 1          | 1          | 1          | 1   |
| Application of Concepts      | 25%           | 1.5        | 1.5        | 1.5        | 1.5        | 1.5 |
| Decision Making              | 20%           | 1.5        | 1.5        | 1.5        | 1.5        | 1.5 |
| Clarity of Response          | 15%           | 1/2        | 1/2        | 1/2        | 1/2        | 1/2 |

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Tharigai S.

## Annexure A

### Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

**2001:0db8:abcd:0000:0000:0000:0000/64**

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24



A packet arrives at the router with the destination IP address:

**192.168.2.45**

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

#### RUBRICS FOR CALCULATION

Scenario Based Assessment

| Criteria                     | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               |
|------------------------------|---------------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Understanding of Scenario    | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20%           | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |
| Application of Concepts      | 25%           | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |
| Decision Making              | 20%           | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |
| Clarity of Response          | 15%           | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |

## 1. Smart Campus network design:

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### Scenario Analysis:

The Smart Campus must divide the given  $192.168.100.0/24$  Network among three departments according to their required subnet sizes while Minimizing IP Wastage.

### Solution:

School of Computing (125)

\* Network Address :  $192.168.100.0$

\* Broadcast Address :  $192.168.100.127$

\* Host Range :  $192.168.100.1 - 192.168.100.126$

\* Usable Hosts : 126

School of electronics (126)

\* Network :  $192.168.100.128$

\* Broadcast :  $192.168.100.191$

\* Host range :  $192.168.100.129 - 192.168.100.190$

\* Usable Hosts : 62

School of Mechanical (127)

\* Network :  $192.168.100.192$

\* Broadcast :  $192.168.100.223$

\* Host Range :  $192.168.100.193 - 192.168.100.222$

\* Usable Hosts : 30



Unused Address Range.

\* 192.168.100.224 - 192.168.100.255.

Decision

The subnet allocation satisfies each departments requirements, reduces broadcasts traffic, avoid IP Wastage and reserves one /27 subnet for future expansion.

## 2. Software company network Allocation.

Scenario Analysis:

The company must allocate subnets for 10.10.0.0 /16 using VLSM so that each department receives sufficient IP addresses without the available address space.

Remaining unused addresses begin from 10.10.1.224 while most of the /16 network remains available for future branches

Departments:

Networks

Usable Hosts.

Software

development

10.10.0.0/24

254.

Quality

Assurance

10.10.10/25.

126.

Human

Resources

10.10.1.128/26

62.

Executive

Managerent

10.10.1.192/27

30.

Decision:

Allocating larger subnets first ensures efficient utilization of the address space, supports organizational growth and simplifies future network management.



### 3. IPv6 Migration

Scenario analysis.

The hospital is migrating to IPv6 to accommodate thousands of IoT devices and medical equipment without running out of IP addresses

Solution:

\* Compressed Address:  $2001:db8:abcd::167$

\* Expand Addresses:  $2001:0db8:abcd:0000:0000:0000:0000:0000$

\* Network prefix (164):  $2001:db8:abcd:0000$

\* Host portion: Last 64 bits.

\* Total Hosts Addresses:  $2^{64} = 18,446,744,073,709,557,616$ .

Decision:

IPv6 provides an exactly large addresses space, removes address exhaustion and is ideal for supporting future healthcare IoT infrastructure with efficient routing.

## 4. ISP Routing Decision.

### Scenario Analysis:

A router receives a packet with destination IP 192.168.2.45. The router must determine the correct route before forwarding the packet.

### Solution:

- \* Destination networks : 192.168.2.0 /24.
- \* Matching Routing Entry : 192.168.2.0 /24.
- \* Next Hop: Interface connected to 192.168.2.0/24
- \* If no matching route exists, use default route (0.0.0.0 /0)

### Decision:

The router forwards the packet through the matching subnet using the longest-prefix match. This ensures accurate and efficient packet delivery while reducing routing errors.

## 5. office LAN configuration.

### scenario Analysis.

Two departments require separate LAN's connected through a layer-3 router while maintaining communication b/w them.

### Solution:

Admin LAN (192.168.0.0 /26)

\* Network : 192.168.10.0

\* Broadcast : 192.168.10.63

\* Host Range : 192.168.10.1 - 192.168.10.62.

\* PCs : 192.168.10.1, 3, 4.

\* Gateway : 192.168.10.1

Finance LAN (192.168.10.64/26)

\* Network : 192.168.10.64

\* Broadcast : 192.168.10.127.

\* Host Range : 192.168.10.65 - 192.168.10.126.

\* PC's : 192.168.10.65 - 192.168.10.126.

\* Gateway : 192.168.10.65.